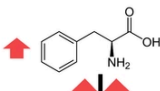
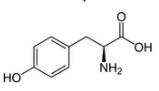


# 5.9 Emerging Treatments

- ▼ Explain what a disease caused by an inborn error of metabolism looks like.
  - This is a disease where there may be a **missing/faulty enzyme** which results in the **build up of the wrong product, excess build up of the right product, or lack of build up of the right product**
- ▼ An example of a disease caused by an inborn error of metabolism.
  - An example is **phenylketonuria (PKU)** where a **lack of Phenylalanine hydroxylase** leads to the **build up of phenylketones**

### Phenylketonuria (PKU)

Phenylalanine → Phenylketones

Phenylalanine hydroxylase

Tyrosine



**Untreated PKU**

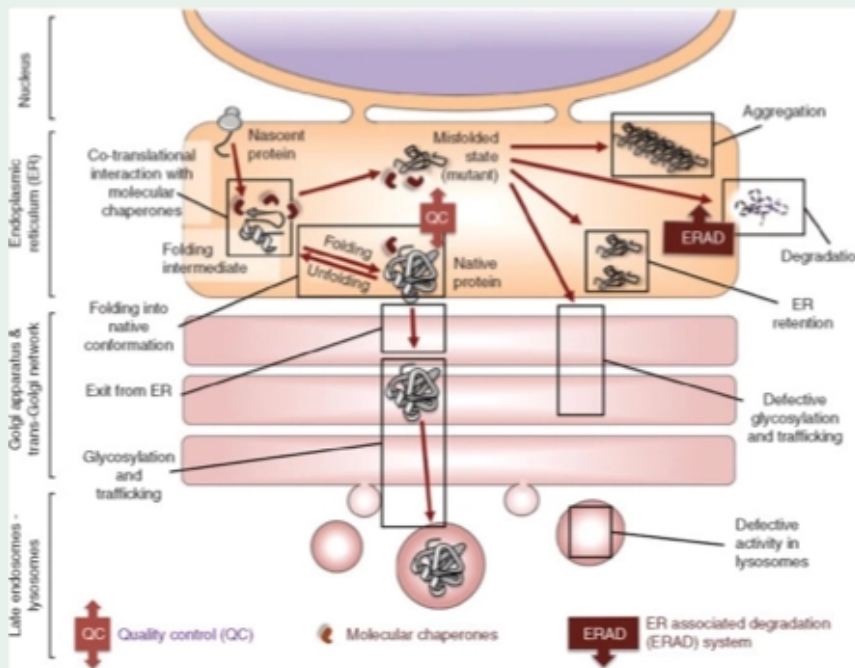
Major cognitive impairment  
behavioural difficulties  
fairer skin, hair and eyes than siblings  
lack of melanin  
recurrent vomiting

Emerging treatments POM 4.9 Explain treatment options for genetic diseases and the utility of novel therapeutic approaches.

- Also, **MCADD deficiencies**
- ▼ How can a disease caused by an error of metabolism be treated?
  - **Dietary alterations**
    - i.e if you have MCADD and so cannot metabolise fatty acids, eat a high carbohydrate diet
- ▼ Explain what a disease caused by missing factors looks like.
  - It will lead to a disease such as **haemophilia** - a blood clotting factor deficiency (Factor VIIIa)
- ▼ How is the problem of infected blood samples combatted in haemophilia treatment?
  - **Heat is used** to treat the infected blood sample and **kill the viruses**
  - The desired gene can then be **cloned and recombined for the patients**
- ▼ How can a disease caused by missing factors be treated?
  - **Inject the patient with the missing factors** they need

▼ Describe how pharmacological chaperones work.

- Proteins and their intermediates are usually synthesised and folded where they can be packaged off to their destination
- However, sometimes (due to mutations) there can be aberrant folding of proteins
- **Pharmacological chaperones** work by **binding onto these aberrantly folded proteins** and **rescuing them** by allowing them to assume a normal shape again.



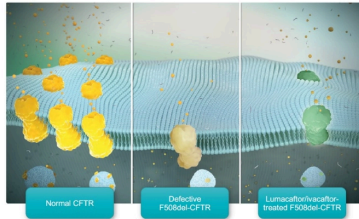
▼ Describe how pharmacological modulators work.

- These are simpler drugs, in that they can have a variety of different effects that **can allow the patient to be less affected by any defects they may have**
- Their effects may be varied, i.e.
  - Receptor agonists/antagonists
  - Ion channel activators/blockers
- For example, **CF causes Chloride ion channel defectiveness**. A pharmacological modulator can activate this ion channel

▼ What is combination therapy?

- This is a method of treatment which involves **combining two drugs together, so their effects can be experienced by the patient simultaneously**

## Combination therapy



### Cystic fibrosis

Defective chloride channel

One mutation (f508del) misfolded, inactive channel

Treat with combination chaperone and activator

Orkambi (Ivacaftor/lumacaftor) – NICE approved Oct '19

Not cure does improve lung function

▼ What is the difference between a mis-sense and a non-sense mutation?

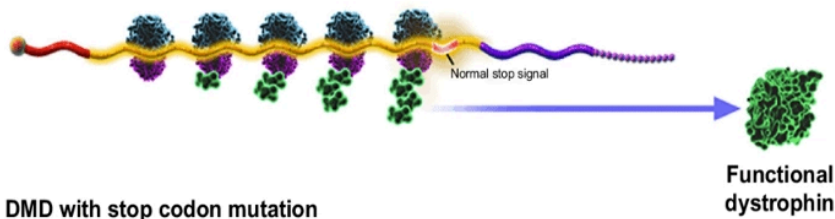
- A **mis-sense mutation** is a more normal mutation where the change in base sequence leads to a different amino acid being produced and hence unusual protein
- A **non-sense mutation** is a more severe mutation where the change in base sequence leads to a stop codon being produced earlier than usual
  - Hence, **no/fewer protein is produced**

▼ How are aminoglycoside antibiotics able to aid non-sense suppression therapy?

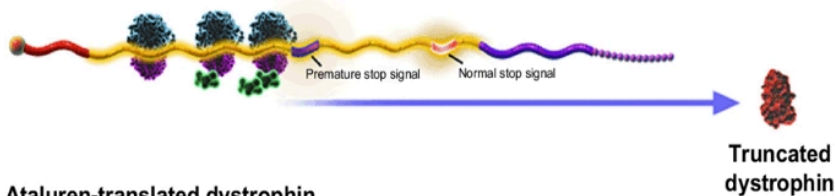
- They can **bind onto ribosomes**, and **replace amino acids that result from mutations** with tRNA sequences that are similar
- This means the ribosome is now able to perform **stop-codon read-through** and **still produce the protein**

▼ NON-SENSE SUPPRESSION IN DUCHENNE MUSCULAR DYSTROPHY.

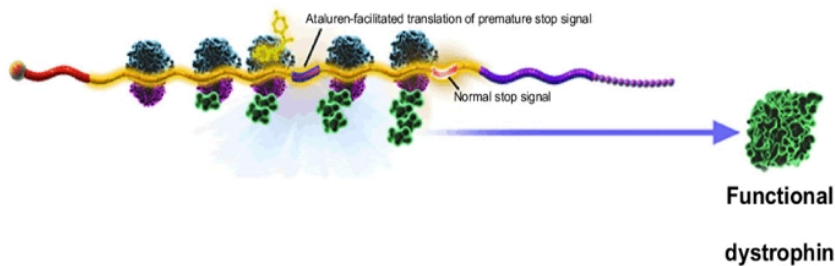
Normal control



DMD with stop codon mutation



Ataluren-translated dystrophin



▼ What do the terms in vivo, ex vivo and in vitro mean?

- **In vivo** - in the living
- **Ex vivo** - out of the living
- **In vitro** - in glass

▼ In gene therapy, how are dominant and recessive diseases dealt with?

- **Dominant genes** are **deleted**
- **Recessive genes** are **altered**

▼ What is a pronucleus?

- The egg nucleus, **once it is haploid**, is called the **female pronucleus**.
- The sperm nucleus, **once it is haploid**, is called the **male pronucleus**.

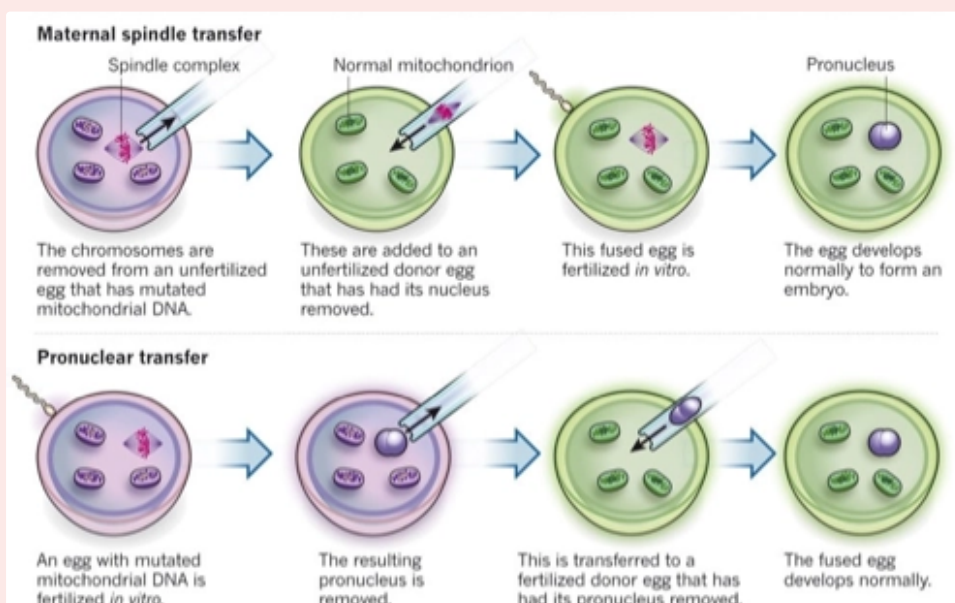
- ▼ Describe the two methods that IVF is used for mitochondrial disease treatment.

### Maternal spindle transfer

- **Chromosomes are removed from an unfertilised egg** that has mutated mitochondrial DNA
- The **chromosomes are added to an unfertilised donor egg** which has healthy mitochondrial DNA
- This fused egg is **fertilised in vitro**
- The egg develops normally to form an embryo
- **TRANSFER THE CHROMOSOMES INTO CLEAN MITOCHONDRIA CELLS TO BE FERTILISED**

### Pronuclear transfer

- The egg with mutated mitochondrial DNA is fertilised in vitro
- The **resulting pronucleus is removed**
- The **pronucleus is transferred to a normal egg cell** with healthy mitochondrial DNA and has had its pronucleus removed
- The **fused egg develops normally**
- **TRANSFER THE FERTILISED PRONUCLEI OUT OF CELLS INTO CLEAN CELLS TO CONTINUE DEVELOPMENT**
- i.e either take chromosomes out and fertilise them in a clean cell. or fertilise the cells first and transfer the pronucleus into the clean cell after



## ▼ When is gene silencing used?

- Against **dominant diseases** that result in a **“gain-of-function” phenotype**

## ▼ What are oligonucleotide base treatments used for?

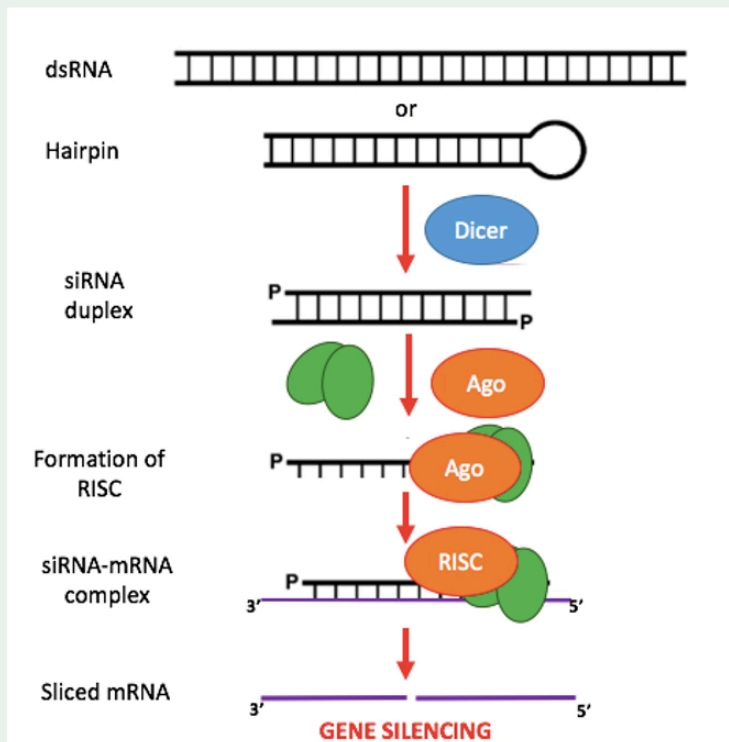
- **Gene silencing**
  - This can be in **dominant diseases (gain-of-function specifically)**, **recessive diseases**, and **non-genetic diseases**.

## ▼ Describe how an antisense oligonucleotide works.

- An **antisense oligonucleotide** works by binding onto its target region of mRNA during protein synthesis and **preventing translation into proteins**
  - The mRNA can also be destroyed once the antisense oligonucleotide (ASO) has bound to it
  - The **ASO** will also typically **bind to the site of pathogenesis**
- It can also **alter splicing**

## ▼ How does siRNA (RNAi) work?

- It **binds to target RNA** and because it is **double-stranded**, **cleaves it**, and **prevents it from functioning**.
- This **prevents protein production** via this gene region



▼ What is the difference between an antisense oligonucleotide and RNAi (siRNA)?

- An **antisense oligonucleotide** is **single-stranded**
- **siRNA** is **double-stranded**

▼ What is viral gene therapy?

- It is a technique that can be used to **engineer viruses to carry therapeutic genes to help patients produce proteins they may lack**

▼ Explain how viral gene therapy works.

- Modified DNA is **injected into a vector**
- **Vector is taken into the host cell via endocytosis and is released**